R&D | RetailPulse

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Power BI Dashboard Requirements

> **In simple words**
>
> This is the blueprint for the dashboard. It is built as a story: each page has a headline that states the insight, and the last page gives the actions.

1. Design idea: the dashboard tells a story

A manager should see the main message in **30 seconds**. So every page has a **headline title written as a sentence** (not just a chart name), a few visuals, and one text box with the takeaway.

Weak chart title	Insight headline (better)
WAPE by Family	[X] families cause [Y]% of total forecast error
Bias by Month	[Month] is the only month where we clearly under-forecast
WAPE by City	Error is high in [N] small cities, not everywhere

The headline can be a **dynamic measure** (a text measure that fills the family name and numbers from the data), so it updates with filters.

2. Data model

- Import the **error fact table** (store x family x date) from PostgreSQL. Add dimension tables: date, store, family.
- Relationships: fact to date, store, family. One direction, one-to-many.
- If the file is too slow, aggregate to store x family x week in SQL and say so in the report.
- Rename columns to readable names (Store Number, not store_nbr). Hide key columns.

3. Global slicers (every page)

Slicer	Field	Style
Date	Year and month	Dropdown or range
Region	State, then city	Hierarchy dropdown
Family	Product family	Dropdown with search
Store type	Type A to E	Buttons

4. Pages

Page 1: Executive Summary	
Headline	Overall accuracy is [X]%. [N] families cause [Y]% of the error.
Manager question	Are we doing well overall? Where is the problem?
Visuals	KPI cards: WAPE, Accuracy %, Bias %, Excess units, Shortfall units. Bar: top 5 families by WAPE. Line: monthly WAPE trend. Text box: 3 key takeaways.
Interaction	Click a family to filter the page. Button to Page 5.

Page 2: Category Deep-Dive	
Headline	[Family list] are the priority: high error and high volume.
Manager question	Which families should we fix first, and in which direction do they miss?
Visuals	Scatter: x = total sales, y = WAPE, one dot per family (top right = priority). Pareto bar + cumulative line. Stacked bar: excess vs shortfall units by family.
Interaction	Drill-through to Family Detail page (daily actual vs forecast).

Page 3: Regional Analysis	
Headline	Error is highest in [states or cities], mostly where volume is [low/high].
Manager question	Is the problem everywhere or in specific places?
Visuals	Map or bar: WAPE by city. Matrix: state > city with WAPE, Bias % and conditional colors. Bar: WAPE by store type.
Interaction	Tooltip: top error family in that city.

Page 4: Seasons and Events	
Headline	[Month] and [event] are the hardest periods. Holiday weeks lean to [excess/shortfall].
Manager question	When does the baseline fail, and which way?
Visuals	Column: WAPE by month. Bar: holiday vs week-after-holiday vs regular. Matrix heatmap: weekday x month. Cards: payday vs other days, promotion vs no promotion.
Interaction	Slicer for named period (Christmas, Carnival, school start).

Page 5: Insights and Actions	
Headline	Five findings and what we recommend.
Manager question	What should we do, who owns it, and how soon?
Visuals	5 insight cards (text boxes): finding, evidence number, recommendation, owner, priority (High, Medium, Low). No new charts. Link each card to the page that proves it.
Interaction	Buttons back to pages 1 to 4.

5. Hidden drill-through page: Family Detail

Line chart of daily actual vs forecast for one family, with holiday days marked. Cards for WAPE, Bias % and top 3 error days. Helps explain why a family is high error.

6. Measures to create (names and logic)

Measure	Logic (plain words)
Total Actual Sales	Sum of actual sales
Total Forecast	Sum of forecast sales
Total Abs Error	Sum of absolute error
WAPE %	DIVIDE(Total Abs Error, Total Actual Sales)
Accuracy %	1 - WAPE %
Bias %	DIVIDE(Total Forecast - Total Actual Sales, Total Actual Sales)
Excess Units	Sum of excess_units
Shortfall Units	Sum of shortfall_units
Error Share %	Abs error of selected group / abs error of all groups
High Error Flag	1 if WAPE % above 30% and sales above the minimum volume
Direction Label	Over if Bias % above +5%, Under if below -5%, else Balanced
Headline Text	Text measure that puts the top family and its numbers into a sentence

Use `DIVIDE` so a zero in the bottom does not crash the measure. Never average a WAPE column. Always use the measures.

7. Design rules

- Maximum 6 visuals per page. One message per visual.
- Colors: gray-blue by default. Red only for high error. Orange for excess, blue for shortfall, and use them the same on every page.
- No 3D, no pie with more than 4 slices, sort bars from high to low.
- Axis and card titles in plain words, units shown (units, %).
- Add a **Reset filters** bookmark and page navigation buttons.
- A small note on each page: data period and baseline used.

8. Dashboard checks before sharing

- Totals on Page 1 match the SQL totals (sales, abs error, WAPE).
- Every slicer changes the numbers it should.
- Every headline is true for the current filters (or says 'all data').
- No visual is empty, cut off or overlapping on the screen.

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